PriceCompare

Android Application

User Manual

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1. Introduction

PriceCompare is an Android application that helps you track, compare, and manage product prices across different stores. Whether you are grocery shopping or comparing electronics, PriceCompare keeps all your price records in one place so you always know where to get the best deal.

2. Getting Started

SYSTEM REQUIREMENTS

- Android 8.0 (Oreo) or higher
- Active internet connection (for account sync and data backup)
- Camera permission (required for barcode scanning and photo capture)

INSTALLATION STEPS

1. Download the PriceCompare APK file to your Android device.
2. Open device Settings and enable "Install from unknown sources" if prompted.
3. Tap the APK file and follow the on-screen installation steps.
4. Once installed, tap the PriceCompare icon on your home screen to launch the app.

3. Login & Registration

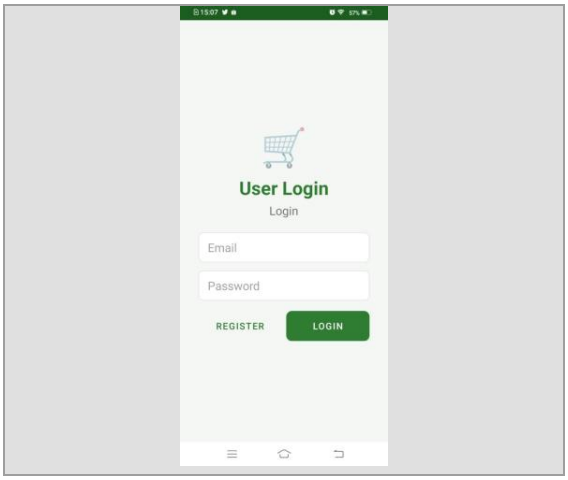


Figure 3.1 Login Screen

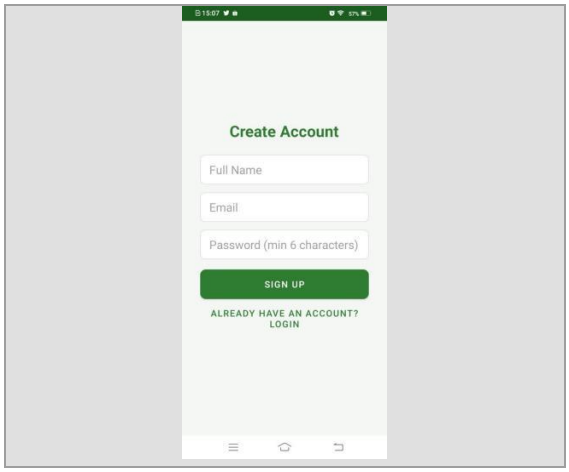


Figure 3.2 Register Screen

3.1 Creating an Account

PURPOSE

Create a free account to start saving and syncing your price records.

STEPS

1. Open the app. The Login screen appears.
2. Tap Register at the bottom of the screen.
3. Enter your Email, Username, and Password.
4. Tap the Register button.

> ✓ **Expected Result**
> You are taken directly to the login and your account is ready to use.

3.2 Logging In

PURPOSE

Sign in to access your saved products and price records.

STEPS

1. Enter your registered Email and Password.
2. Tap Login.

> ✓ **Expected Result**
> The Dashboard opens, showing all your tracked products.

> ⚠ **Important**
> Your password must be at least 6 characters long. Use a strong password to keep your data safe.

4. Dashboard & Profile

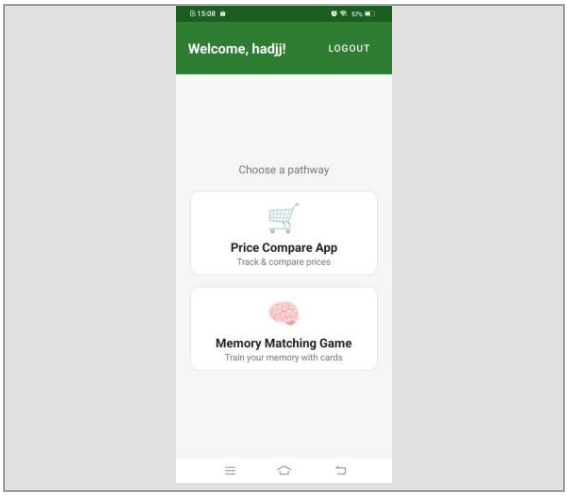


Figure 4.1 Dashboard Screen

4.1 Dashboard

PURPOSE

After logging in, the Dashboard appears. From here you can access Price Compare App and Memory Matching Card Game.

NAVIGATION TILES

- Price Compare
  Manage and browse your tracked products.
- Memory Matching Card Game
  Play the memory card game.

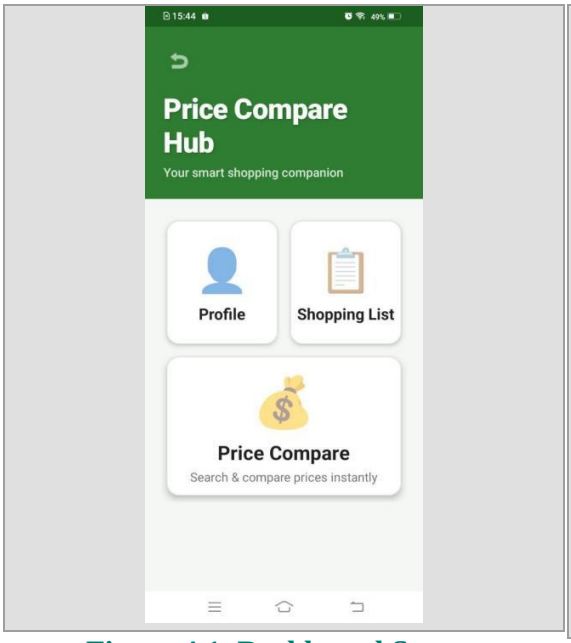


Figure 4.1 Dashboard Screen

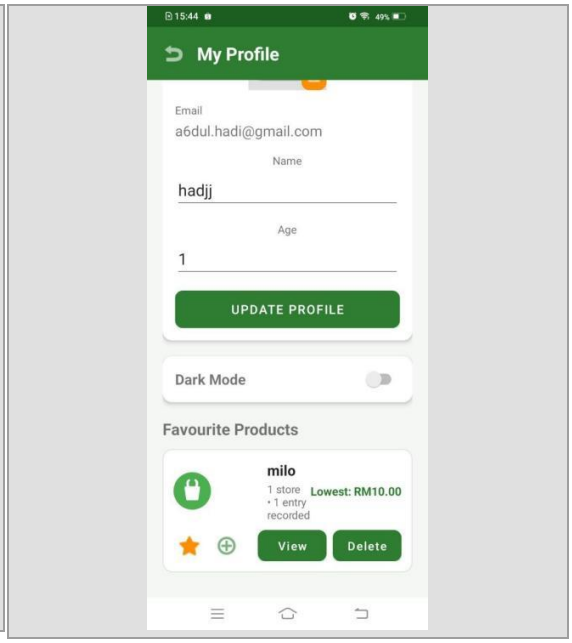


Figure 4.2 Profile Screen

4.2 Price Compare Dashboard & Profile

PURPOSE

View your account information, select the price compare, or shopping list.

STEPS — UPDATING YOUR PROFILE

1. Tap the Price Compare app
2. Tap the Profile icon on the Dashboard .
3. Set your name,age to update your profile
4. Dark Mode is available
5. Favourite Product also available at My Profile

5. Price Compare — Product List

PURPOSE

The Product List shows all the products you are currently tracking. Use it to browse, search, filter, and manage your price entries.

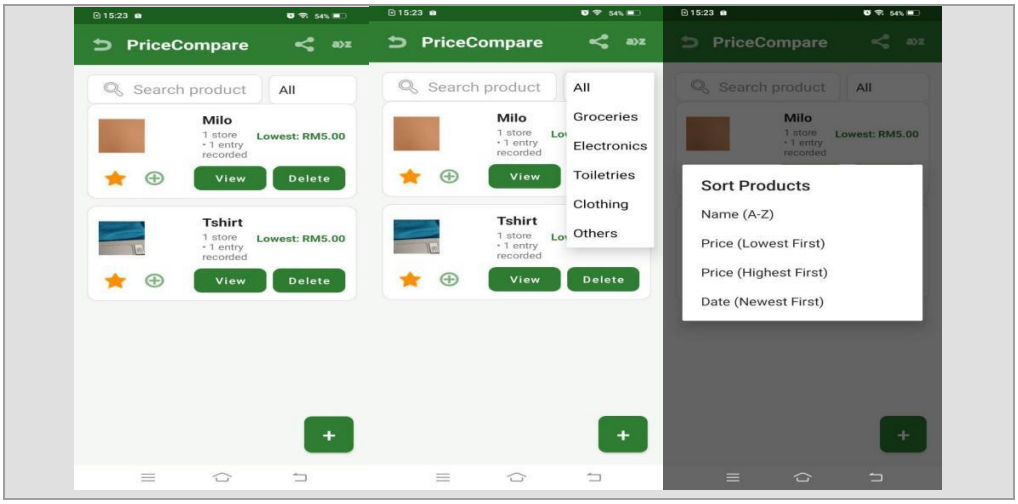


Figure 5.1 Product List Screen

Reading a Product Card

- Category icon (or product photo, if one has been taken)
- Product name
- Number of stores and price entries recorded
- Lowest recorded price — shown in green

Toolbar Options

- Search bar — type to filter products by name.
- Category filter — choose from All, Groceries, Electronics, Toiletries, Clothing, or Others.
- A–Z button — sorts the list alphabetically.
- Share button — share your product list with others.

Card Action Buttons

- View — opens the Product Detail page.
- Delete — permanently removes the product and all its price records.
- ★ (Star) — marks the product as a favourite.
- ✚(Plus circle) — adds the product to your Shopping List.

> ⚠ **Important**
> Tapping Delete on a product card removes all associated price records permanently. This action cannot be undone.

6. Adding a Product

PURPOSE

Record a new price entry for a product at a specific store.

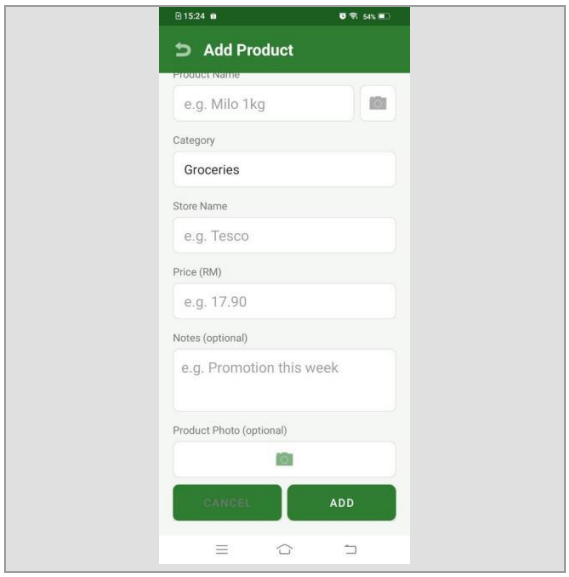


Figure 6.1 Add Product Screen

STEPS

1. On the Product List screen, tap the green + button (bottom-right).
   The Add Product form opens.
2. Enter the Product Name.
   Auto-complete suggestions appear for existing product names — select one to avoid duplicates.
3. Choose a Category from the dropdown (Groceries, Electronics, Toiletries, Clothing, Others).
4. Enter the Store Name.
   Auto-complete shows stores you have used before.
5. Enter the Price (RM), e.g. 17.90.
6. Add any Notes (optional), e.g. "Promotion this week".
7. Tap Add to save the entry, or Cancel to discard.

✓ Expected Result

The new price entry is saved and the product appears in your Product List.

Scanning a Barcode

STEPS

1. Tap the camera icon next to the Product Name field.
2. Point the camera at the product barcode.
3. The app detects the barcode and fills in the product details automatically.

> **Tip**
> Use the barcode scanner to speed up product entry and avoid typos in product names.

Taking a Product Photo

STEPS

1. Tap the Product Photo button on the Add Product form.
2. Take a photo of the product.
3. A preview appears below the button. The photo is linked to this price entry.

> **Tip**
> Photos are stored on your device only and are not uploaded online. They will not appear on other devices.

7. Product Detail Page

PURPOSE

View all price records for a single product, grouped by store. Compare prices, view a price trend graph, and manage individual entries.

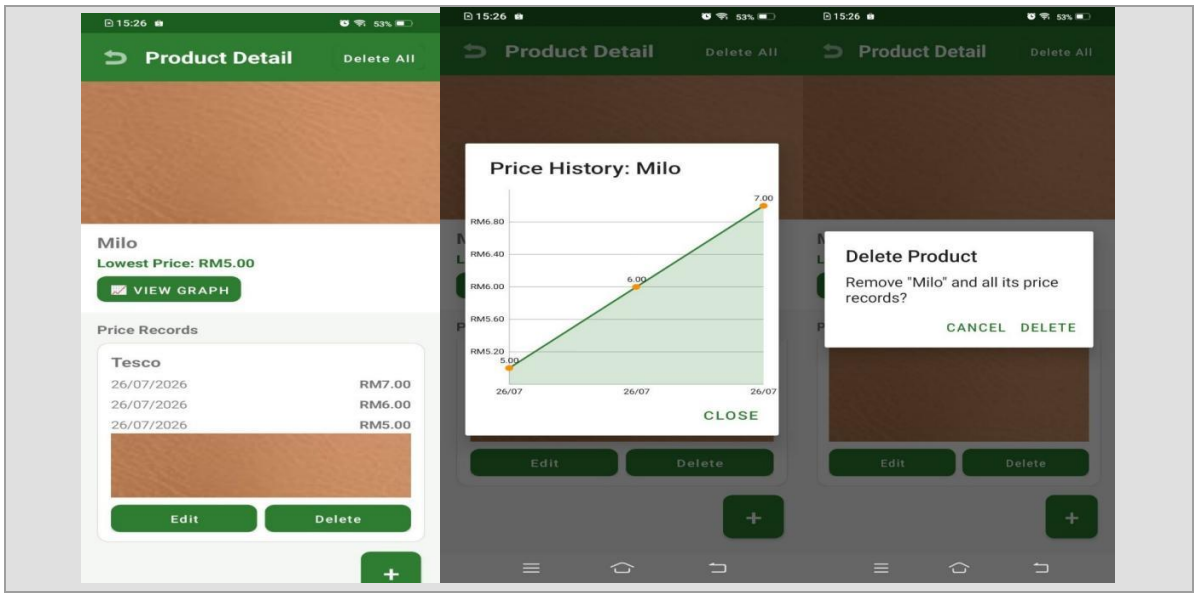


Figure 7.1 Product Detail Screen

STEPS — OPENING THE DETAIL PAGE

1. On the Product List, tap View on any product card.

Expected Result

The Product Detail page opens, showing all price records grouped by store.

Reading the Page

- Product photo (full width, top) — shown if any entry has a photo.
- Product name and lowest recorded price (highlighted in green).
- View Graph button — appears when 2 or more entries exist.
- □ note icon — indicates that a note was added to the entry.

Viewing the Price Graph

STEPS

1. Tap View Graph on the Product Detail page.
2. A line chart appears showing price over time for all entries.
3. Pinch to zoom or drag to scroll through the graph.
4. Tap Close to dismiss the graph.

Deleting All Records for a Product

STEPS

1. Tap Delete All (top-right of the Product Detail page).
2. Confirm the deletion when prompted.

> ⚠ **Important**
> Deleting all records permanently removes the product and every price entry associated with it. This cannot be undone.

8. Editing a Price Record & Shopping List

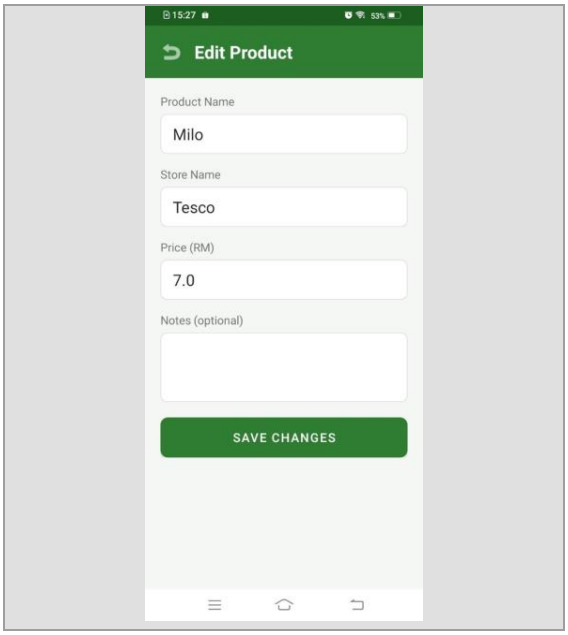


Figure 8.1 Edit Product Screen

8.1 Editing a Price Record

PURPOSE

Update the most recent price entry for a store — for example when a price changes or you made a typo.

STEPS

1. On the Product Detail page, tap Edit on the relevant store card.
   The edit form opens, pre-filled with the current values.
2. Update the Store Name, Price, or Notes as needed.
3. Tap Save to apply changes, or Cancel to discard.

✓ Expected Result

The store card on the Product Detail page reflects the updated price immediately.

Tip

Editing only changes the most recent entry for that store. Older entries for the same store remain unchanged.

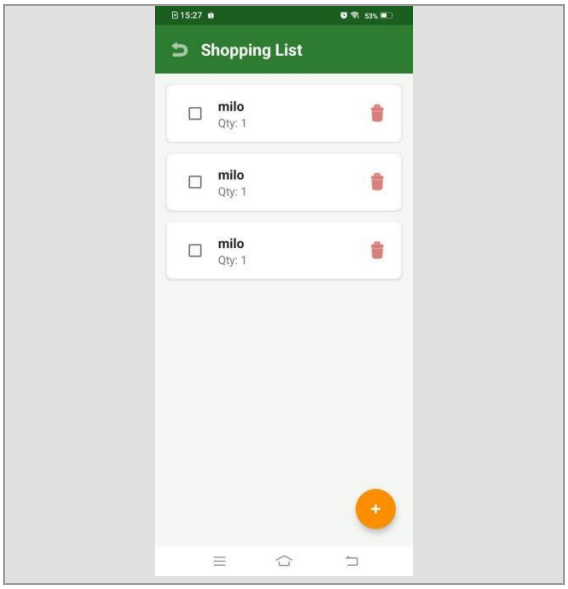


Figure 8.2 Shopping List

8.2 Shopping List

PURPOSE

Keep a checklist of items you need to buy on your next shopping trip.

STEPS — ADDING AN ITEM

1. Tap the + button on the Shopping List screen.
2. Type the item name and tap Add.

> **Tip**
> You can also add any tracked product to the Shopping List by tapping the + icon on its card in the Product List.

STEPS — CHECKING OFF AN ITEM

1. Go back to the Dashboard and select Shopping List
2. Tap an item to mark it as purchased.
   The item shows a strikethrough to indicate it is done.
3. Tap again to uncheck it if needed.

STEPS — REMOVING AN ITEM

1. Tap the Delete button next to the item you want to remove.

> **Expected Result**
> The item is removed from your Shopping List immediately.

9. Memory Matching Card Game

PURPOSE

The Memory Matching Card Game challenges you to find all matching pairs of cards before running out of time or moves, depending on your chosen mode. Cards are displayed face-down in a grid; tap two at a time to reveal them. Matched pairs stay revealed; mismatched cards flip back face-down. Complete all pairs to win.

9.1 Game Hub & Navigation

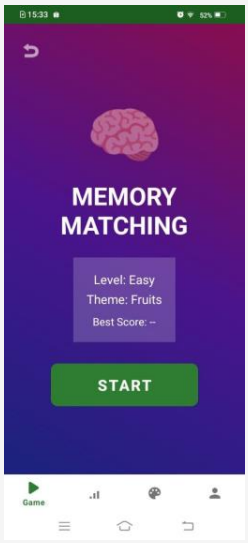


Figure 9.1 Game Hub Screen

When you tap Memory Matching Card Game on the Dashboard, the Game Hub opens. It has four tabs at the bottom:

- Game — summary of current settings and the Start Game button.
- Level — choose difficulty level and game mode.
- Theme — choose card symbol theme, card colour, dark mode, and audio settings.
- Profile — view your game statistics and match history.

9.2 Choosing a Level & Game Mode

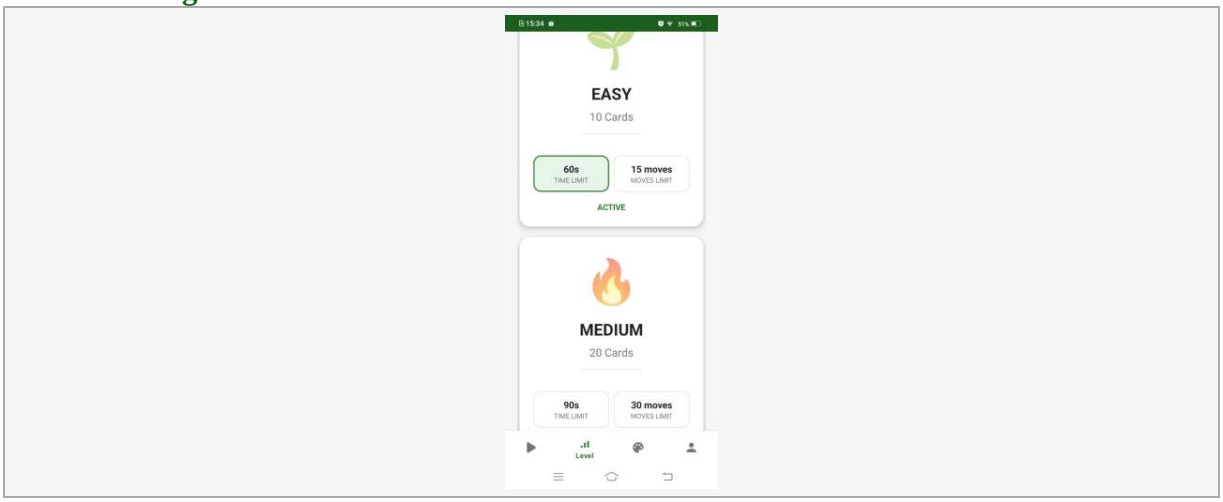


Figure 9.2 Level Selection Screen

Tap the Level tab to choose your difficulty and game mode.

DIFFICULTY LEVELS

Level	Details
Easy	10 cards (5 pairs) in a 2 × 5 grid. Option Time Limit: 60 s | Moves Limit: 15 moves.
Medium	20 cards (10 pairs) in a 4 × 5 grid. Option Time Limit: 90 s | Moves Limit: 30 moves.
Hard	30 cards (15 pairs) in a 5 × 6 grid. OptionTime Limit: 120 s | Moves Limit: 45 moves.

GAME MODES

• Time Limit — match all pairs before the countdown timer reaches zero. A progress bar shows the remaining time; it turns red when 10 seconds remain.

• Moves Limit — match all pairs within the allowed number of moves. Each time you flip a second card (whether it matches or not) counts as one move. The moves counter turns red when 5 or fewer moves remain.

Tap Time or Moves under any level card to activate that combination. The selected combination is highlighted and labelled ACTIVE.

9.3 Choosing a Card Theme

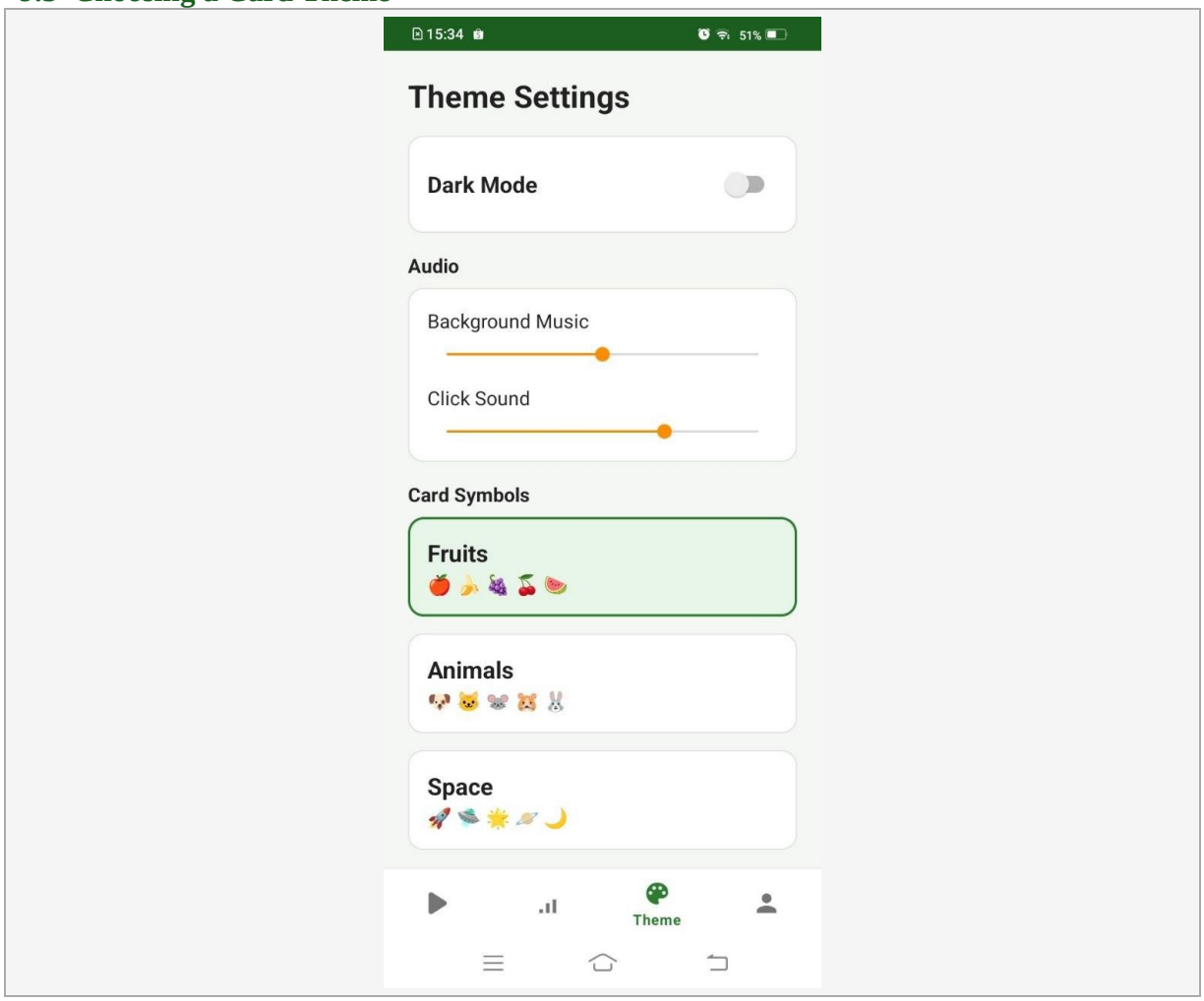


Figure 9.3 Theme Selection Screen

Tap the Theme tab to customise the look and feel of the game.

CARD SYMBOL THEMES

Theme	Symbols used on cards
Fruits	□□□□□□□□□□□□□□□
Animals	□□□□□□□□□□□□□□□
Space	□□□□□ ☌□□□□□ ⭑□□□
Ocean	□□□□□□□□□□□□□ ⚓□□

CARD COLOUR

Tap any of the five colour swatches — Green, Orange, Purple, Teal, or Pink — to change the colour of face-down cards. The change takes effect in the next game.

DISPLAY & AUDIO

- Dark Mode — toggle the switch to enable or disable dark mode. The app restarts its theme immediately.
- Background Music — drag the slider to adjust background music volume (0 – 100).
- Click Sound — drag the slider to adjust sound-effect volume for card flips, matches, and mismatches (0 – 100).

9.4 Playing a Game

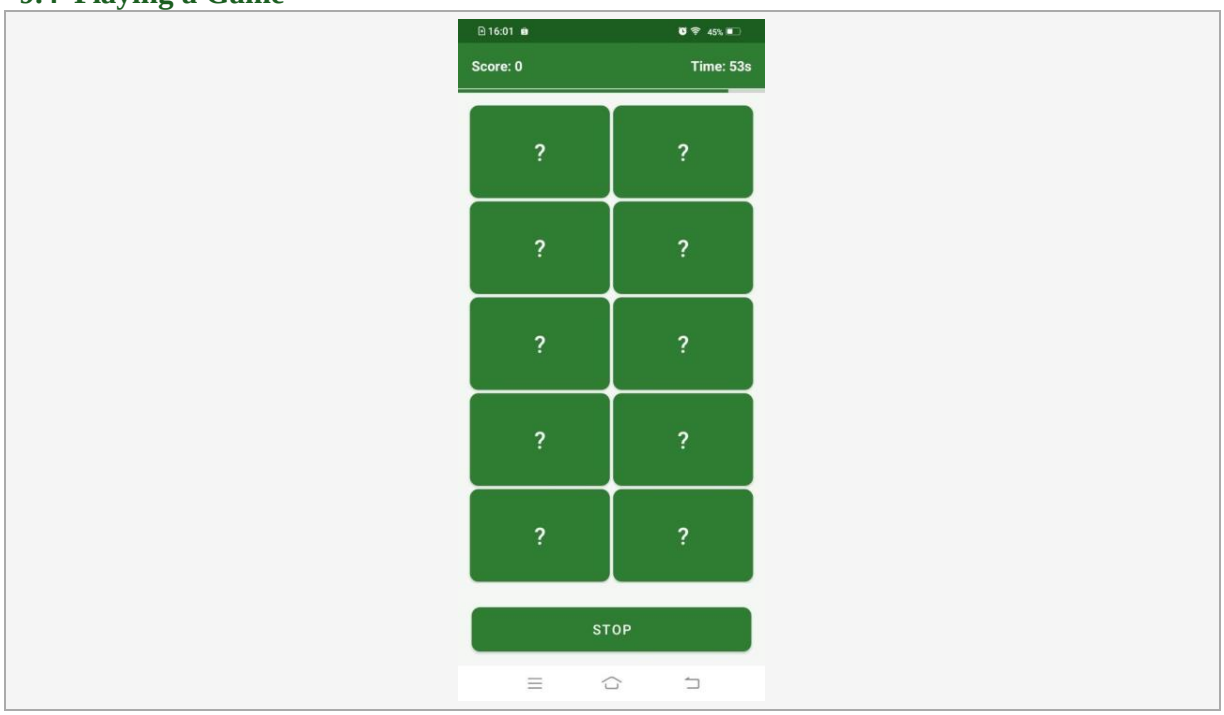


Figure 9.4 Game Screen — cards face-down, score and timer visible

STEPS — STARTING A GAME

1. From any tab in the Game Hub, tap the Game tab.
2. Confirm your chosen Level, Mode, and Theme are shown in the summary.
3. Tap Start Game. All cards appear face-down with a "?" symbol.
4. Tap any card to reveal its symbol — a flip animation plays and a flip sound is heard.
5. Tap a second card to attempt a match.
   - Match — both cards pulse and stay revealed; your score increases by 10 points.
   - Mismatch — both cards shake and flip back face-down after a short delay.
6. Continue until all pairs are matched (win) or the timer/moves run out (lose).

✓ **Expected Result** When all pairs are matched the game ends immediately and the Result screen appears, even if time or moves remain.

DURING THE GAME

- Score — shown at the top. Each matched pair earns +10 points.
- Timer (Time Limit mode) — a countdown and a progress bar show remaining seconds.
- Moves Remaining (Moves Limit mode) — displayed at the top-right.
- Stop button — tap to end the game early. The Result screen opens and a loss is recorded.

Tip In Moves Limit mode, only the second card tap counts as a move — use your first tap to explore without spending a move.

9.5 Game Result Screen

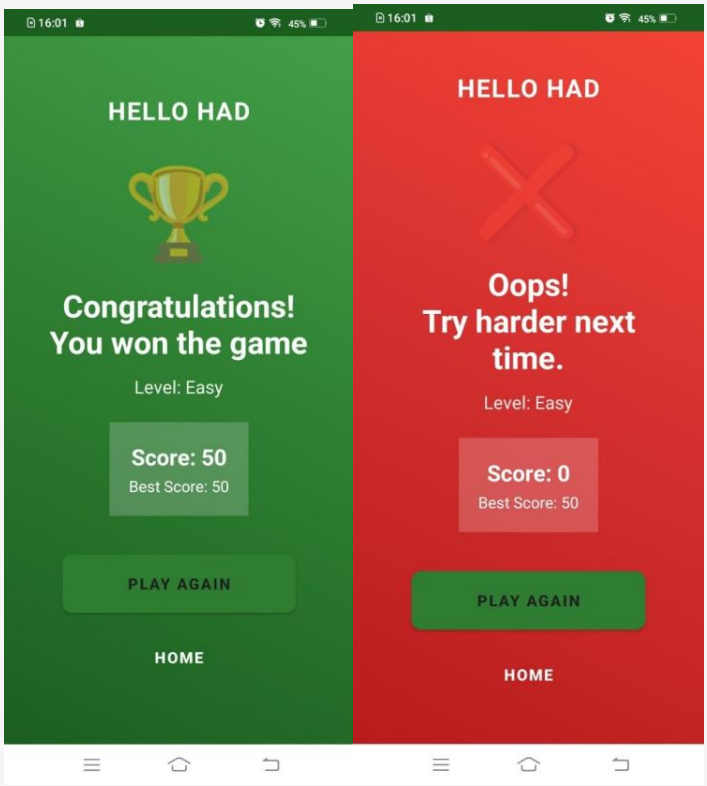


Figure 9.5 Result Screen — Win state (left) and Lose state (right)

After the game ends, the Result screen appears with an animated entry showing:

- Player greeting (your profile name or Firebase display name).
- □ Congratulations! (win) or ✕Oops! Try harder next time. (lose).
- Level — the difficulty level played.
- Score — points earned in this session (10 per matched pair).
- Best Score — your all-time highest score for this level.

Two buttons are available:

- Play Again — immediately starts a new game with the same settings.
- Home — returns to the Game Hub.

9.6 Game Profile & History

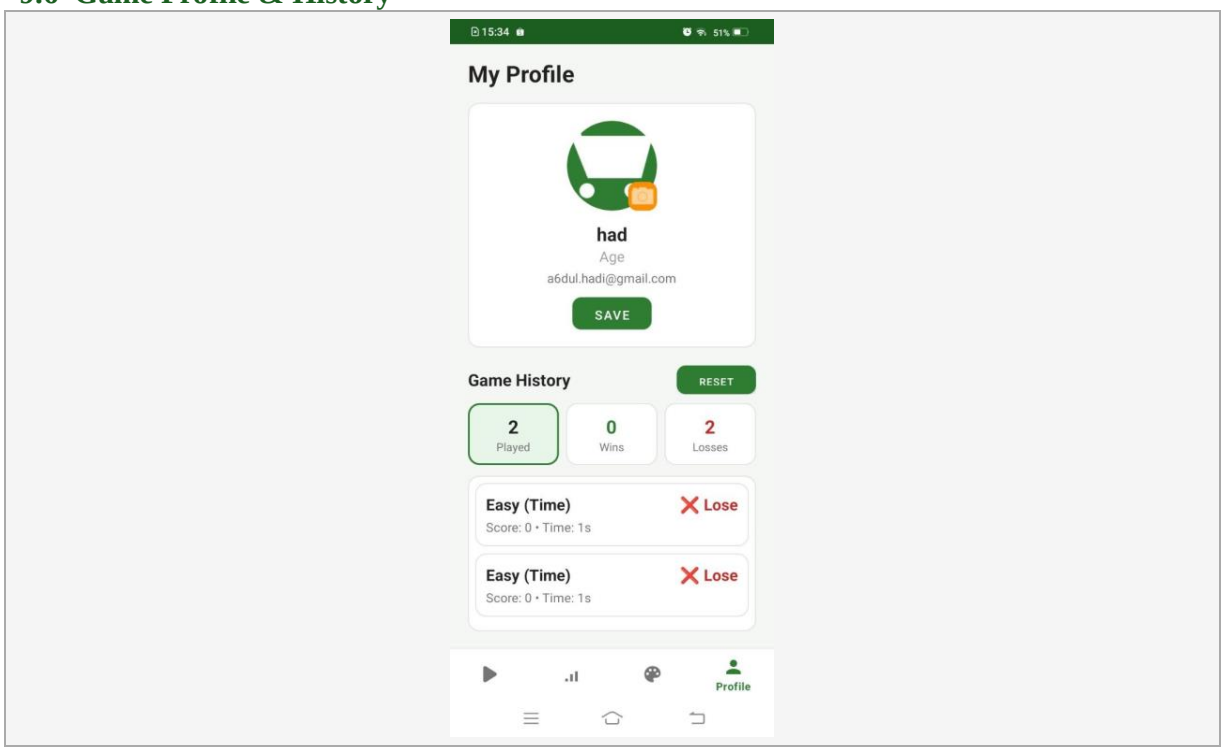


Figure 9.6 Game Profile Screen

Tap the Profile tab in the Game Hub to view and manage your game profile.

PROFILE DETAILS

- Profile photo — tap the pencil icon to choose a photo from your device gallery.
- Name and Age — edit and tap Save Profile to persist the changes.
- Email — your registered account email (read-only).
- Games Played, Games Won, Games Lost — lifetime statistics.

MATCH HISTORY

The history list shows your last 50 games, each entry displaying:

- Level and mode played (e.g. "Easy (Time)").
- □ Win or ✕Lose result.
- Score and time elapsed.

Use the filter buttons to show All, Wins only, or Losses only.

RESETTING HISTORY

1. Tap Reset History at the bottom of the Profile tab.
2. All history, wins, and losses counters are cleared immediately.

⚠**Important** Resetting history is permanent and cannot be undone. High scores are also cleared.

10. Troubleshooting

If you encounter an issue, refer to the table below for quick solutions.

Problem	Solution
Cannot log in	Check that your email and password are correct. Passwords are case-sensitive.
Camera does not open	Go to device Settings → Apps → PriceCompare → Permissions and enable Camera.
Barcode not detected	Ensure good lighting and hold the camera steady. Clean the camera lens if needed.
Product photo not showing on another device	Photos are stored locally on the device where they were taken and are not synced online.
No internet / data not syncing	Check your Wi-Fi or mobile data connection and try again.
Data not updating after a change	Close and reopen the app to refresh your data.
App crashes or freezes	Restart the app. If the problem persists, reinstall PriceCompare from the Play Store.
Deleted a product by mistake	Deletions are permanent and cannot be recovered. Always confirm before deleting.

11. Frequently Asked Questions

Why does my photo not appear on another device?
Photos are saved locally on your device and are not synced online. They will only appear on the device where the photo was taken.

Can I add the same product from multiple stores?
Yes. Use the exact same product name ,auto-complete helps ensure consistency and choose a different store name each time. All entries will be grouped under one product on the Product Detail page.

How do I find which store has the best price?
Open the Product Detail page for the product. The entry with the lowest price is marked with a ★ star, and the lowest price is shown in green at the top of the page.

The barcode scanner is not working. What should I do?
Make sure Camera permission is granted to PriceCompare. Go to device Settings → Apps → PriceCompare → Permissions and enable Camera. Also ensure there is adequate lighting when scanning.

How do I delete a single price entry (not the whole product)?
On the Product Detail page, tap Delete on the relevant store card. If that store has multiple entries, a list appears ,then tap the specific entry you want to remove.

I accidentally deleted a product. Can I recover it?
No. Deletions are permanent and cannot be reversed. Always double-check before confirming a deletion.

Is my data backed up online?
Yes. Your account information and price records are securely stored online and will be available when you log in on any device. Note that product photos are the exception as they are stored on your device only.